Transformation from inactive provitamin D to the active form is accomplished by exposure to sunlight (UV rays). Cholecalciferol is converted to calcifediol (25-hydroxy cholecalciferol) in the liver. Similarly, Ergocalciferol is converted to 25-hydroxyergocalciferol. These calicifediols are further hydroxylated to form calcitriols (1,25-dihydroxy cholecalciferol & 1,25-dihydroxy ergocalciferol), the biologically active form of vitamin D. The calcitriol circulates as a hormone in the blood.

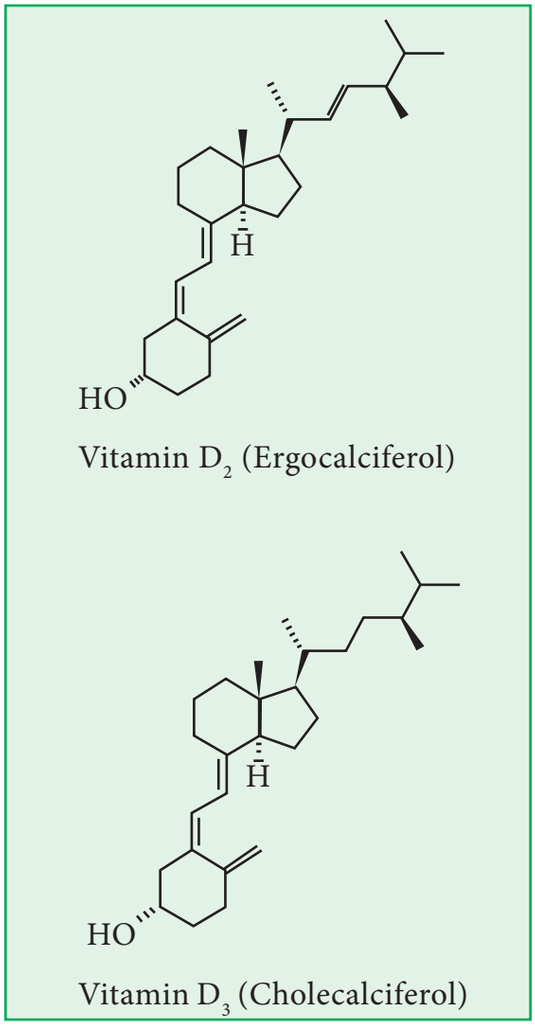

Figure 8.4 Structure of vitamin D

Sources:

Fish liver oil is the richest source of vitamin D. Milk, butter and egg yolk also contain considerable quantity of vitamin D (Fig. 8.5).

Egg yolks: 3- 10 microgram/g

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